

Lap2 : Python

Calculate Area Function.

```
def calc_area(shape, s1, s2=None):
    shape = shape.lower()
    if shape == 't':
        return 0.5 * s1 * s2
    elif shape == 'r':
        if s2 == None:
            return s1 * s1
        else:
            return s1 * s2
    elif shape == 'c':
        return math.pi * (s1 ** 2)
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/019_Python/Solution/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/019_Python/Solution/Lab2$ python3 script.py
Triangle (10, 7): 35.0
Rectangle (10, 7): 70
Square (10): 100
Circle (10): 314.16
nagy@ZenBook-UX425EA:~/Mine/ITI/019_Python/Solution/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/019_Python/Solution/Lab2$
```

Name List to Dictionary.

```
def names_to_dict(names_list):
    names_list.sort()
    res = {}
    for name in names_list:
        key = name[0].lower()
        if key not in res:
            res[key] = []
        res[key].append(name)
    return res
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/019_Python/Solution/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/019_Python/Solution/Lab2$ python3 script.py
names: ['ahmed', 'fatma', 'Ibrahim', 'faten']
{'i': ['Ibrahim'], 'a': ['ahmed'], 'f': ['faten', 'fatma']}
nagy@ZenBook-UX425EA:~/Mine/ITI/019_Python/Solution/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/019_Python/Solution/Lab2$
```

Mario Pyramid.

```
def mario_pyramid(height):  
    for i in range(1, height + 1):  
        spaces = " " * (height - i)  
        stars = "*" * i  
        print(f"{spaces}{stars}")
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/019_Python/Solution/Lab2$  
nagy@ZenBook-UX425EA:~/Mine/ITI/019_Python/Solution/Lab2$ python3 script.py  
*  
**  
***  
****  
    *  
   **  
  ***  
 ****  
*****  
*****  
*****  
*****  
nagy@ZenBook-UX425EA:~/Mine/ITI/019_Python/Solution/Lab2$  
nagy@ZenBook-UX425EA:~/Mine/ITI/019_Python/Solution/Lab2$
```